Basic Principles of Medicine 2
Module: Neuroscience and Behaviour

Clinical Neuroimaging:

Directed Learning Activity

To be studied after Lecture Organization of the Nervous System

Images from *Neuroanatomy: An Atlas of Structures, Sections and Systems 10th Ed. D.E. Haines © 2019 Lippincott Williams and Wilkins* unless otherwise stated

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Recommended Reading

1. Siegel and Sapru, Essential Neuroscience, Fourth Edition, 2019, Chapter 27, Vascular Syndromes.

Part II:
Computed Tomography (CT-scan)

To be studied after Lecture
Organization of the Nervous
System

Learning Objectives

SOM.CS.I.BPM2.1.NSCI.0048 Describe the physics, diagnostic use and contraindications/limitations of X-ray-based techniques (excluding CT).

SOM.CS.I.BPM2.1.NSCI.0049 Describe the physics, diagnostic use and contraindications/limitations of CT.

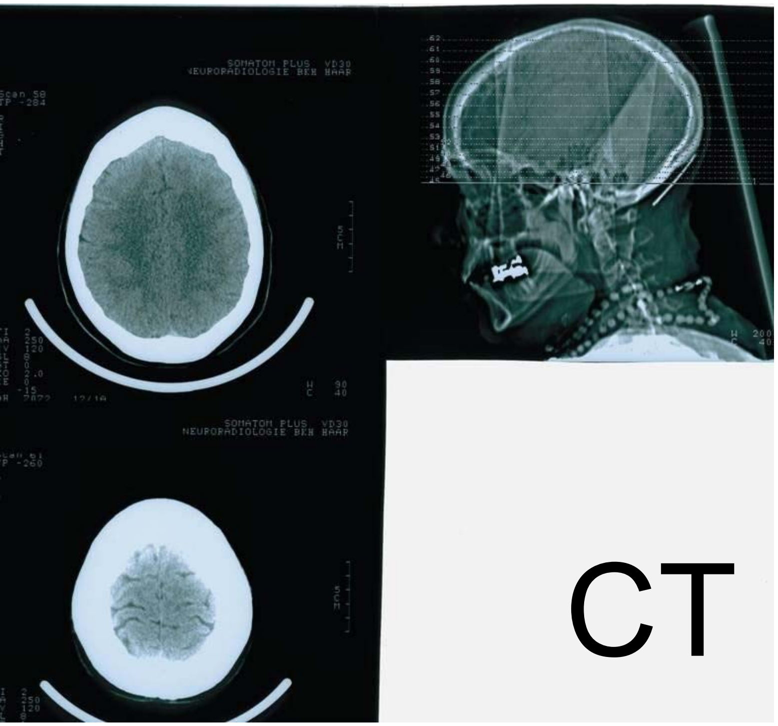
SOM.CS.I.BPM2.1.NSCI.0050 Describe the physics, diagnostic use and contraindications/limitations of MRI.

Technique

- In cranial CT scan, an X-ray source rotates around the head of the patient. X-ray sensors located opposite to the source continuously measure the attenuation of the X-radiation. The data are used to calculate a horizontal ‘slice’ (or tome) of the head.

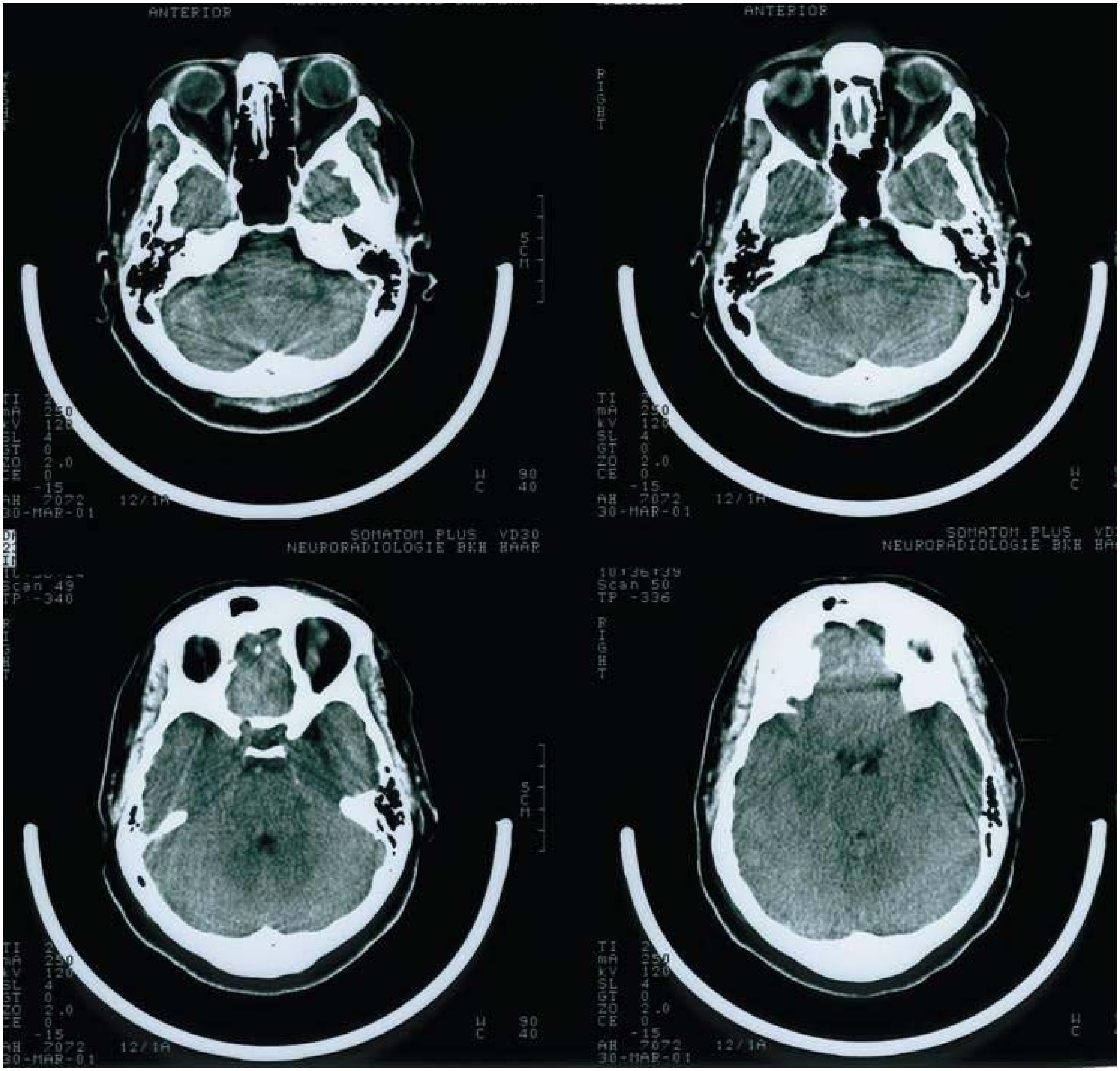
Indications

- Infarctions and particularly recent intracranial hemorrhages are readily detected by CT. The procedure is initially performed without contrast agents, because contrast may resemble a bleed. A normal CT generally does not show an infarction in the acute stage but is commonly performed to exclude a hemorrhage
- For diagnosing tumors, CT with contrast can be used inexpensively but is less sensitive than magnetic resonance imaging (MRI)
- Other indications for CT scan are increased intracranial pressure (before lumbar puncture) and head trauma with facial or skull fractures

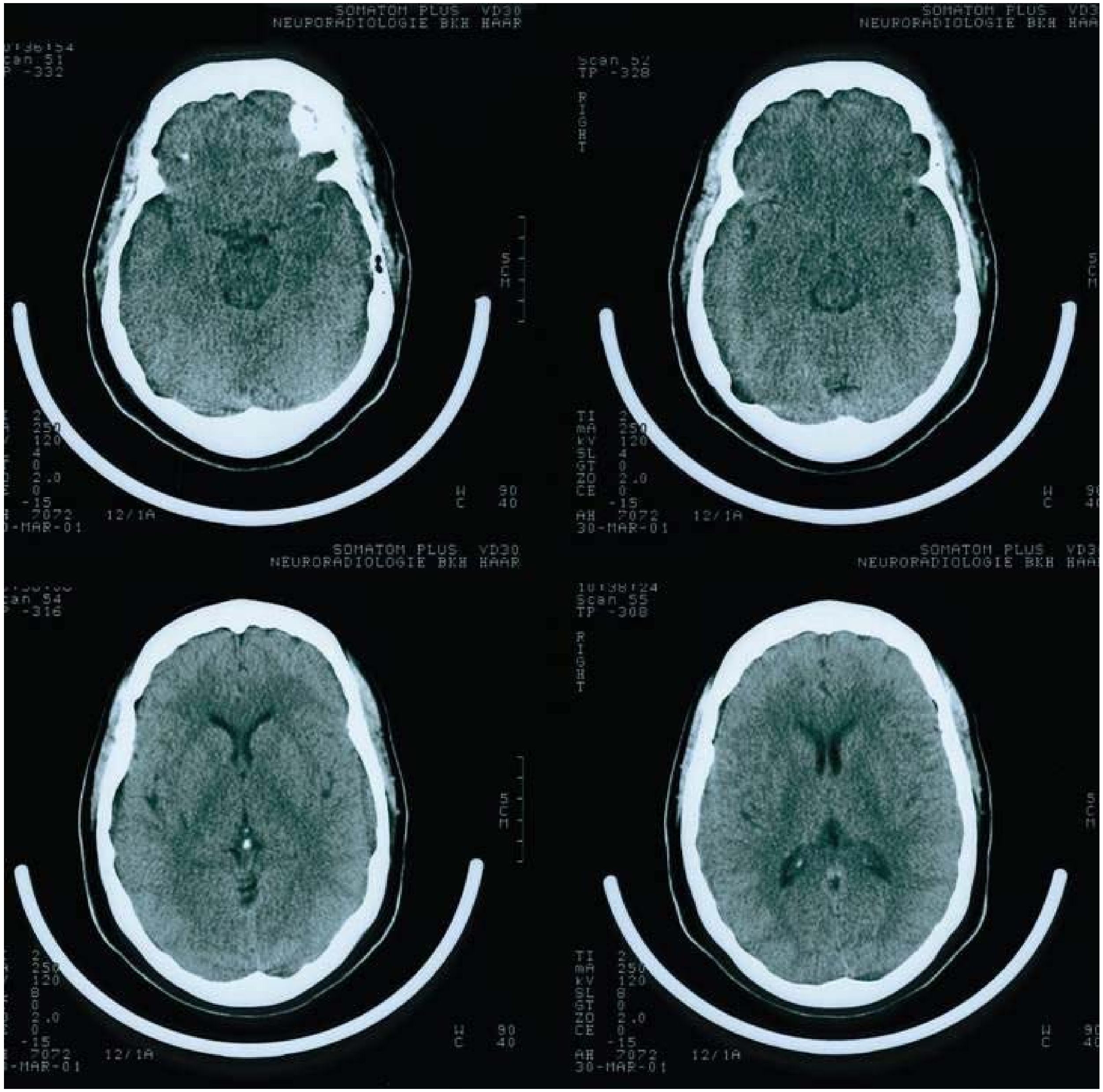


http://upload.wikimedia.org/wikipedia/commons/1/13/Rosies_ct_scan.jpg

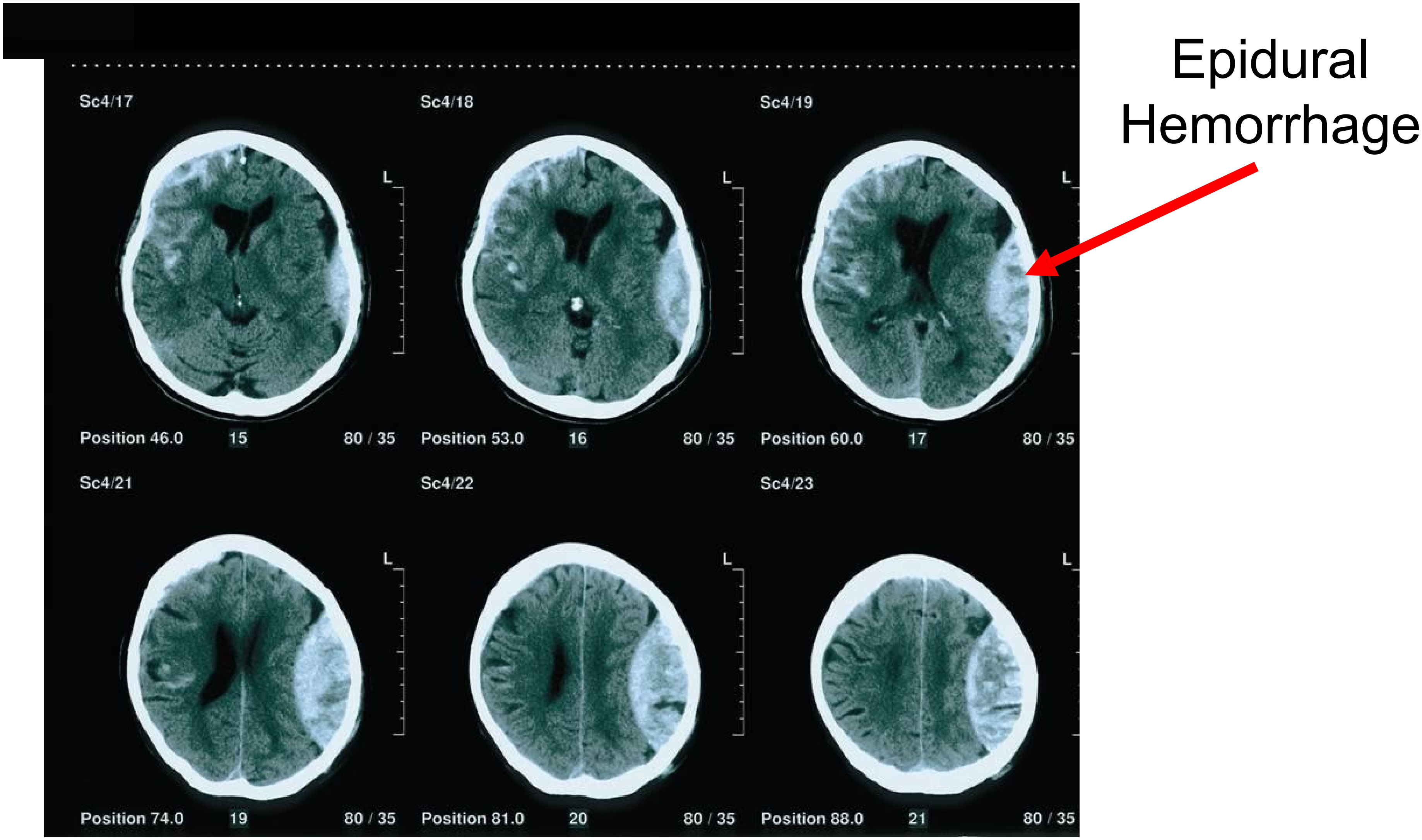
- X-rays pass from a rotating source around the head (or other body part), with unabsorbed X-rays reaching a rotating detector
- After a rotation, the X-ray source and detectors are moved to a new plane, creating a new slice (tome)
- CT can differentiate bone, white matter, gray matter and fresh blood



CT may suboptimally represent brainstem structures



Note differentiation of white and gray matter in CT

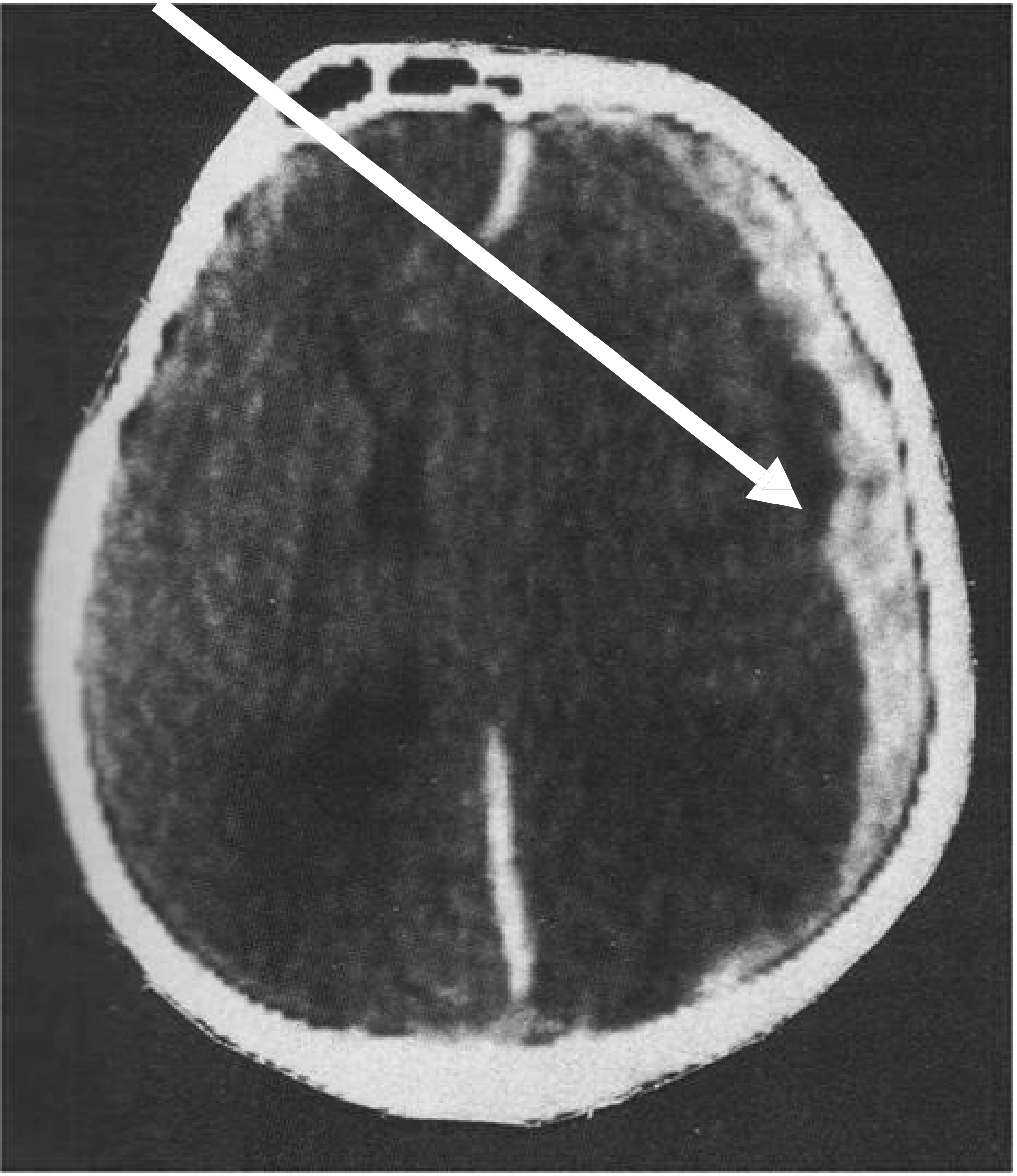


- CT reveals fresh blood, as in this left-sided epidural hemorrhage
- Skull fracture may damage dural vessels, yielding a biconvex accumulation of blood

Subarachnoid Hemorrhage

- Blood in the subarachnoid space, as revealed in CT.

Subdural hemorrhage



- CT often reveals subdural hemorrhages, which often reflect trauma-induced tearing of the dura or tiny cerebral veins that traverse the dura from the subarachnoid space
- Subdural hemorrhages often assume a crescent shape

CT-Angiogram

CT Advantages and Disadvantages

- Advantages
  - Relatively inexpensive
  - Commonly available
  - Sensitive to fresh blood
  - Sensitive to skull fracture and dense foreign matter
  - Entire body can be quickly investigated
- Disadvantages
  - Exposure to ionizing radiation